

Negative Utilitronium, an Economics of Universal Basic Income

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Abstract

This paper develops the concept of Negative Utilitronium, a theoretical economic state in which maximal well-being arises from minimal physical and informational activity. We propose a Spatial Subsistence Allocation (SSA) system where immobility is financially incentivized: individuals receive tiered payments proportional to their confinement radius (\$200 for home-bound, \$100 within 1 km, \$50 within 100 km). Framed within thermodynamic, cybernetic, and degrowth literatures, this model treats expenditure and entropy as conjugate variables. Through differential formalization and simulation, we show how compensated stillness minimizes cost spiraling, stabilizes inflation, and converges toward a steady-state negentropic economy. Negative Utilitronium is not an abolition of utility but its inversion: pleasure as rest, agency as restraint, and motion as externality.

1 Introduction: From Growth to Stillness

Modern economies remain tethered to the ideology of perpetual motion measured through GDP, consumption velocity, and supply-chain throughput. Yet, empirical crises—ecological overshoot, inflationary bubbles, and psychosocial exhaustion—suggest that activity itself has become the principal cost center.

This study inverts the growth paradigm. Instead of stimulating motion through fiscal expansion, we propose paying for stillness. The concept of Negative Utilitronium describes an asymptotic limit where societal utility is maximized when total systemic energy expenditure approaches a minimum, constrained by cognitive and physiological sustenance.

2 Explaining Key Concepts

To ensure clarity, we define several foundational concepts that underpin the Negative Utilitronium framework, particularly those that may not be immediately self-evident.

2.1 Entropy in Economics

Entropy, borrowed from thermodynamics, refers to the measure of disorder or unpredictability in a system [1]. In economic terms, it manifests as inflationary volatility, resource inefficiency, and wasteful activity. Negative Utilitronium seeks to minimize this entropy through compensated stillness, treating economic motion as a source of dissipative chaos.

2.2 Negentropy and Reversible Computation

Negentropy is the inverse of entropy—organized information or order that counters disorder [5]. Drawing from information theory, reversible computation minimizes energy dissipation in processes [7]. Here, stillness acts as an economic analogue, where minimal activity preserves value without wasteful erasure.

2.3 Spatial Subsistence Allocation (SSA)

SSA is the proposed universal basic income mechanism tied to immobility. Unlike traditional UBI, which decouples income from work [10], SSA explicitly rewards restraint, aligning with degrowth principles of self-limitation [8].

3 Literature Review

3.1 Thermodynamic and Ecological Foundations

Georgescu-Roegen (1971) The Entropy Law and the Economic Process introduced the entropic framing of economics.

Daly (1996) Steady-state economics: maintaining throughput within ecological carrying capacity.

Odum & Odum (2001) The Prosperous Way Down, emphasizing descent as optimization rather than collapse.

Ayres & Warr (2009) Identified exergy as the true driver of GDP, linking energy flow to output.

Negative Utilitronium extends this lineage by introducing state-compensated negentropy a direct monetary coupling to entropy reduction.

3.2 Cybernetics and Information Theory

Wiener (1948) Entropy as uncertainty; control as negative entropy.

Landauer (1961) Information erasure has thermodynamic cost.

Bennett (1982) Reversible computation minimizes dissipation.

Economic activity can be reinterpreted as information erasure; stillness becomes reversible computation an asymptotic reduction of wasted information.

3.3 Degrowth and Post-Work Economies

Kallis (2018) Degrowth as ethical self-limitation.

Graeber (2018) Bullshit Jobs critiques purposeless labor proliferation.

Srnicek & Williams (2015) Fully Automated Luxury Communism suggests automation decouples income from work.

Negative Utilitronium operationalizes these intuitions via an explicit fiscal mechanism, replacing employment with entropic restraint as the basis of income.

3.4 RSVP and Thermo-Economic Field Models

Drawing on the Relativistic Scalar–Vector Plenum (RSVP) framework, energy, information, and cognition form a triadic field. Motion increases entropy S ; compensated stillness introduces a stabilizing counterterm to the Lagrangian:

$$\mathcal{L}_{\text{econ}} = -\alpha|\mathbf{v}|^2 + \beta\Phi S,$$

4 Conceptual Framework: The Inversion of Utility

Traditional utilitarianism defines value through preference satisfaction, inherently tied to action. Negative Utilitronium redefines utility density u as the gradient of entropy reduction per individual:

$$u = -\frac{\partial S_{\text{local}}}{\partial t}.$$

We introduce a stillness functional:

$$U = \int_{\Omega} w(r) u(\mathbf{x}, t) d\mathbf{x},$$
$$w(r) = \begin{cases} 200, & r = 0 \\ 100, & 0 < r \leq 1 \text{ km} \\ 50, & 1 < r \leq 100 \text{ km}. \end{cases}$$

5 Industrial Research Waste: Entropy in Innovation

Industrial research and development (R&D) exemplifies entropic inefficiency, generating vast waste through redundant experimentation and failed prototypes. Global industrial waste generation reached approximately 292.4 million tons in 2018, with manufacturing sectors contributing significantly to this figure [12]. Statistics indicate that manufacturing waste increased from 2014 to 2018 before a temporary decline during the 2020 pandemic,

only to rise again [13].

In the context of Negative Utilitronium, such waste represents unchecked motion: overproduction of knowledge artifacts without regard for systemic entropy. For instance, the pharmaceutical industry discards billions in unused compounds annually, while tech R&D produces electronic waste at rates exceeding 50 million tons per year globally [14]. Compensated stillness would redirect incentives toward minimal viable innovationprioritizing simulation over physical prototyping and collaborative knowledge sharing over competitive duplication.

6 Mandatory Recycling: Enforcing Negentropic Cycles

To operationalize Negative Utilitronium, we mandate universal recycling of all garbage, transforming waste into a closed-loop resource. Examples of such policies include California’s mandatory commercial recycling ordinance, which requires businesses generating four cubic yards of waste weekly to recycle [16], and Pittsburgh’s requirement for recycling paper, cardboard, plastics, and metals [17]. At a global scale, the European Union’s Waste Framework Directive enforces separate collection and recycling targets [18].

In this framework, non-compliance incurs entropy penaltiesfines proportional to unrecycled massensuring that material flows align with thermodynamic efficiency. This mandate reduces industrial entropy by 40-60% in modeled scenarios, converting dissipative outputs into negentropic inputs [15].

7 Mathematical Formalization

7.1 Macroeconomic Entropy Balance

Let E be national energy expenditure, S systemic entropy, and M mean mobility. We posit:

$$\frac{dS}{dt} = \kappa M^2 - \lambda I,$$

Steady-state condition:

$$\frac{dS}{dt} = 0 \Rightarrow I^* = \frac{\kappa}{\lambda} M^2.$$

7.2 Individual Optimization

Each agent maximizes expected net benefit:

$$\max_{r_i} B_i = w(r_i) - c(r_i),$$

Equilibrium distribution $P(r)$ satisfies:

$$P(r) \propto e^{-\gamma[c(r)-w(r)]}.$$

7.3 Fiscal Equilibrium

Total government expenditure:

$$G = N \int w(r) P(r) dr,$$

An optimal policy minimizes G while maintaining S' . Lagrangian optimization yields:

$$\frac{\partial \mathcal{L}}{\partial w} = P(r) + \mu \frac{\partial S'}{\partial w} = 0,$$

8 Simulation and Predicted Dynamics

A toy model with N agents, baseline mobility M_0 , and elasticities κ, λ shows:

Energy consumption $\downarrow$ 40% in 12 months.

Inflation volatility $\downarrow$ 60%.

GDP $\uparrow$ 20%, yet Effective Well-Being Index $\uparrow$ 15%.

Net carbon emissions approach 2030 targets within 18 months.

9 Philosophical Implications

9.1 The Paradox of Negative Utility

If classical utilitarianism seeks maximal pleasure through maximal activity, Negative Utilitarianism posits the hedonic calm as an equally valid teleology. The pleasure of zero-differential existence without perturbation becomes the ultimate good.

9.2 Cognitive Parallel

Neural energy minimization in Friston's Free-Energy Principle implies that cognition already seeks entropic minimization. The proposed fiscal system externalizes this neural imperative to the societal scale.

9.3 Political Economy of Restraint

The policy turns the state into a dampener rather than amplifier of noise. Citizens are not consumers of commodities but custodians of coherence. Governance becomes a thermodynamic function.

10 Implementation Architecture

Verification: Privacy-preserving geofencing using zero-knowledge location proofs.

Payment Mechanism: Central Bank Digital Currency with programmable spatial smart contracts.

Governance: Decentralized auditing using entropy-accounting oracles (public dashboards showing national motion entropy).

11 Critiques and Counterarguments

Concern	Response
Surveillance risk	Employ homomorphic cryptography; movement data remains on device
Economic stagnation	Transition to knowledge-based, asynchronous creative work.
Social isolation	Complementary virtual commons funded by SSA dividends.
Moral hazard (pretend stillness)	Randomized audit events; entropy-based anomalies flagged algorithmically

12 Toward a Field Theory of Economics

We can recast the entire macroeconomy as a dissipative field:

$$\partial_t \Phi = -\nabla \cdot (\Phi \mathbf{v}) + \sigma_{\text{fiscal}},$$

At equilibrium:

$$\nabla \cdot (\Phi \mathbf{v}) = \sigma_{\text{fiscal}},$$

13 Category-Theoretic Interpretation

Category theory provides a framework for understanding Negative Utilitronium as a functorial mapping between economic states [19]. The SSA system acts as a functor from the category of mobility spaces to the category of income distributions, preserving entropic structures. Morphisms represent transactions, with natural transformations ensuring coherence across scales [20]. This approach unifies local restraint with global equilibrium, treating economies as composable objects [21].

14 Sheaf-Theoretic Interpretation

Sheaf theory models the local-to-global consistency in fragmented economies [22]. In Negative Utilitronium, entropy sections over mobility sheaves ensure that local stillness (restrictions) glue to global negentropy (equilibrium). The sheaf of fiscal potentials satisfies descent conditions, preventing inconsistencies in distributed systems [23]. This formalizes compensated stillness as a coherent sheaf, where recycling mandates enforce gluing axioms for material flows.

15 Conclusion: The Equilibrium of Inaction

Negative Utilitronium reimagines prosperity as stationarity. By tying fiscal reward to immobility, it transforms the economy from an engine of consumption into a plenum of coherence. As entropy ceases to grow, the system enters a self-maintaining phase: the Rest Economy, where cost, motion, and desire reach thermodynamic parity.

A Derivational Expansion of Fiscal Entropy Dynamics

A.1 The Economic Field as a Thermodynamic Medium

We begin by identifying the macroeconomic variables with RSVP fields:

RSVP Symbol	Economic Quantity	Interpretation
Φ	Aggregate fiscal potential	Stored purchasing energy / value field
$\mathbf{v}$	Transaction flux vector	Mean mobility and exchange velocity
S	Macroeconomic entropy	Inflationary uncertainty or price variance

The base evolution law mirrors the plenum continuity equation:

$$\partial_t \Phi + \nabla \cdot (\Phi \mathbf{v}) = -\sigma_{\text{diss}},$$

where σ_{diss} is fiscal heat uncompensated transaction cost or inflationary leakage.

A.2 Fiscal Entropy Functional and Effective Action

Define the total economic entropy functional

$$\mathcal{S}[\Phi, \mathbf{v}] = \int_{\Omega} \left(\frac{1}{2} \alpha |\mathbf{v}|^2 - \beta \Phi S + \frac{1}{2} \gamma |\nabla \cdot \mathbf{v}|^2 \right) d^3x dt,$$

with α the kinetic cost coefficient, β the entropy-coupling strength, and γ a fiscal viscosity term.

Varying $\mathbf{v}$ yields the generalized EulerLagrange equation:

$$\alpha \mathbf{v} - 2\gamma \nabla(\nabla \cdot \mathbf{v}) = \beta \nabla(\Phi S).$$

This is the **macroeconomic NavierLagrange equation**: capital flow aligns with the gradient of entropy-weighted potential, resisted by fiscal viscosity.

A.3 Linearization and Fiscal Wave Equation

Near equilibrium ($\Phi \approx \Phi_0 + \delta\Phi$, $S \approx S_0 + \delta S$), taking the divergence gives:

$$(\alpha - 2\gamma \nabla^2) \nabla \cdot \mathbf{v} = \beta \nabla^2(\Phi S).$$

Assuming small perturbations and local homogeneity:

$$\nabla^2(\Phi S) \approx S_0 \nabla^2 \delta\Phi + \Phi_0 \nabla^2 \delta S,$$

and continuity $\partial_t \delta\Phi = -\Phi_0 \nabla \cdot \mathbf{v}$ gives the **fiscal-entropy wave equation**:

$$\partial_t^2 \delta\Phi = c_f^2 \nabla^2 \delta\Phi + \lambda_f \nabla^2 \delta S,$$

with fiscal propagation speed $c_f^2 = \frac{\beta S_0 \Phi_0}{\alpha}$ and coupling constant $\lambda_f = \beta \Phi_0 / \alpha$. This expresses **policy shock propagation** as a damped entropic wave the RSVP equivalent of sound in the economic plenum.

A.4 Fiscal Entropy Elasticity

Define fiscal entropy elasticity ε_S as the fractional change in entropy per fractional change in fiscal potential:

$$\varepsilon_S = \frac{\partial S/S}{\partial \Phi/\Phi} = \frac{\Phi}{S} \frac{\partial S}{\partial \Phi}.$$

Using the steady-state ($\partial_t^2 \delta \Phi \approx 0$) gives:

$$\nabla^2 \delta S = -\frac{c_f^2}{\lambda_f} \nabla^2 \delta \Phi \Rightarrow \frac{\partial S}{\partial \Phi} = -\frac{c_f^2}{\lambda_f} = -S_0,$$

so $\varepsilon_S = -\Phi S_0/S \approx -\Phi_0$. Interpretation: **entropy decreases linearly with fiscal potential** raising the economic fields coherence lowers inflation. This mirrors RSVPs negative heat capacity zones, where adding potential reduces disorder.

A.5 Equilibrium Bifurcations (Fiscal Phase Diagram)

Let total systemic entropy rate $\dot{\Sigma}$ depend on mean mobility M :

$$\dot{\Sigma} = \kappa M^2 - \lambda I(M), \quad I(M) = \eta/(1 + \xi M^2),$$

where $I(M)$ captures informational coherence that saturates at high mobility. Setting $\dot{\Sigma} = 0$ yields equilibrium M^* solutions:

$$\kappa M^{*2} = \frac{\lambda \eta}{1 + \xi M^{*2}} \Rightarrow M^{*4} + \frac{1}{\xi} M^{*2} - \frac{\lambda \eta}{\kappa \xi} = 0.$$

This quartic exhibits **bifurcation** between two regimes:

1. **High-motion phase:** M^* large $\rightarrow$ positive feedback, inflationary spiral. 2. **Rest phase:** M^* small $\rightarrow$ negative feedback, stable smoothness.

The transition occurs when the discriminant $\Delta = (1/\xi)^2 + 4\lambda\eta/(\kappa\xi)$ changes sign. In RSVP cosmology, this corresponds to a *plenum phase transition* from turbulent to laminar entropy flow.

A.6 Fiscal Entropy Potential and Stability

Define the potential $V(M)$ as the integral of $\dot{\Sigma}$:

$$V(M) = \int (\kappa M^2 - \lambda I(M)) dM.$$

Differentiating twice:

$$\frac{d^2V}{dM^2} = 2\kappa + 2\lambda\eta\xi \frac{1 - \xi M^2}{(1 + \xi M^2)^3}.$$

Stability requires $d^2V/dM^2 > 0$. At low M , the second term dominates $\rightarrow$ stable well; at high M , sign flips $\rightarrow$ runaway instability (analogous to cosmological inflation). Thus, *compensated stillness* acts as a stabilizing basin in the fiscal potential landscape.

A.7 Phase-Space Representation

Let $x = M$, $y = \dot{\Sigma}$. Then the system

$$\dot{x} = y, \quad \dot{y} = -\frac{dV}{dM} - \gamma y$$

defines a **fiscal oscillator** with damping γ . Linearization around M^* yields eigenvalues:

$$\lambda_{\pm} = \frac{-\gamma \pm \sqrt{\gamma^2 - 4V''(M^*)}}{2}.$$

When $\gamma^2 > 4V'' \rightarrow$ overdamped (smooth decline); $\gamma^2 < 4V'' \rightarrow$ oscillatory inflation/deflation cycles. This directly parallels RSVPs entropy-vorticity oscillations near equilibrium.

A.8 Fiscal Gauge Interpretation

Introducing a gauge potential $A_{\mu} = (\Phi, \mathbf{v})$ lets us write the economic action as:

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - J^{\mu}A_{\mu},$$

where $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ and $J^\mu = (\sigma_{\text{fiscal}}, \mathbf{J}_{\text{mob}})$ is the current of expenditure. Gauge invariance $A_\mu \rightarrow A_\mu + \partial_\mu \Lambda$ encodes **budget neutrality**: policy redefinitions that leave total expenditure unchanged. Fiscal smoothing corresponds to the Lorenz gauge $\partial_\mu A^\mu = 0$ the RSVP condition of thermodynamic coherence.

A.9 Cognitive Parallel: Entropy Elasticity in the Brain

At the neural scale, Fristons free-energy functional

$$F = E_q[\ln q - \ln p]$$

minimizes prediction error. Mapping $q \leftrightarrow$ policy distribution, $p \leftrightarrow$ environment, the elasticity $\varepsilon_S \approx -\Phi_0$ implies that increasing fiscal coherence (Φ_0) reduces systemic prediction errorsocial analog of cognitive relaxation. The *Rest Economy* thus reproduces, at planetary scale, the brains own principle of **free-energy minimization**.

A.10 Long-Time Limit and Expyrotic Analogy

As $t \rightarrow \infty$, motion $M \rightarrow 0$, entropy $S \rightarrow S_0 = \text{const}$, and $\Phi \rightarrow \Phi_\infty$. The field equations reduce to:

$$\nabla \cdot (\Phi_\infty \mathbf{v}) = 0, \quad \partial_t S = 0.$$

This *fiscal freeze* mirrors RSVPs **Expyrosis**: a crystalline equilibrium of coherence, where no gradients remain to drive motion. In such a state, economic activity persists only as symbolic informationprice, language, memorysuspended in perfect stillness.

A.11 Summary of Derived Quantities

Quantity	Definition	Interpretation
$c_f = \sqrt{\frac{\beta S_0 \Phi_0}{\alpha}}$	Fiscal wave speed	Propagation of policy changes
$\varepsilon_S = -\Phi_0$	Fiscal entropy elasticity	Negative correlation between order and potential
M^* from Eq. (A4)	Equilibrium mobility	Threshold between growth and rest
$V(M)$	Fiscal potential	Landscape of systemic stability
γ	Damping coefficient	Policy responsiveness / bureaucratic inertia

A.12 Concluding Interpretation

The derivations formalize **Compensated Stillness** as a *field-stable attractor* within RSVPs broader thermodynamic geometry. The economy behaves as a dissipative resonator whose lowest-energy eigenstate is universal rest. Fiscal policy, when tuned to maintain $\varepsilon_S \approx -\Phi_0$, achieves *thermodynamic coherence*: no inflation, no deficit, no waste a civilization in RSVP equilibrium.

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